EE 354/CE 361/MATH 310 -

Introduction to Probability

and Statistics

Spring 2026

EE 354/CE 361/MATH 310 L4 section (Spring 2026)

Introduction to Probability and Statistics -

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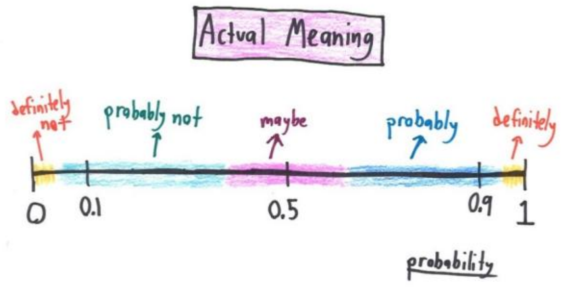
Week 12 - Lecture No 23 – April 01, 2026



Announcements!

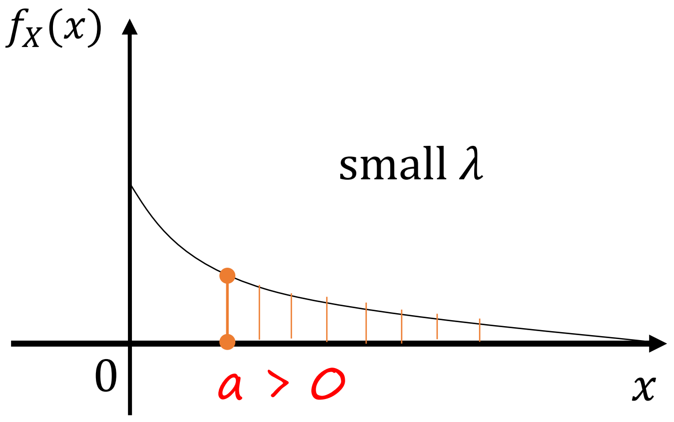
1. Office hours (post Ramadan) – Tuesday & Thursday 12:00 to 1:00 PM
2. **Planned quizzes**
    1. Quiz No 07 – April 08, 2026 (Wednesday)
    2. Quiz No 08 - April 15, 2026 (Wednesday)
    3. Quiz No 09 - April 22, 2026 (Wednesday)
    4. Quiz No 10 - April 29, 2026 (Wednesday)

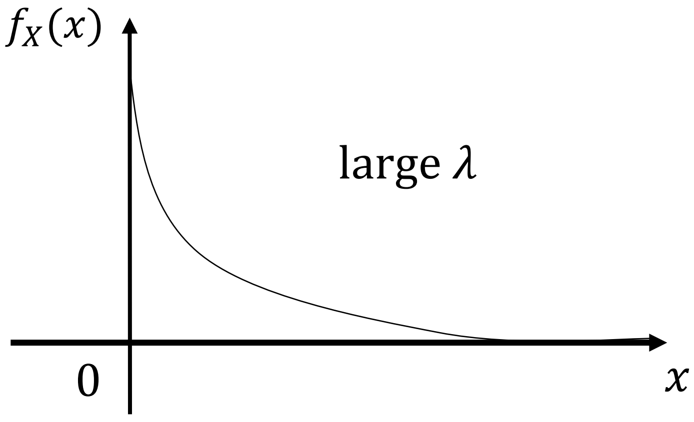
Agenda for today

1. **Quiz No 06**
2. **Unit 5 – Continuous Random Variable**

Recap - Exponential Random Variable; parameter $\lambda > 0$

$$f_X(x) = \begin{cases} \lambda e^{-\lambda x}, & x \geq 0 \\ 0, & x < 0 \end{cases}$$





$P(X \geq a) = ?$

$\Rightarrow$ $P(X \geq a) = \int_a^{+\infty} \lambda e^{-\lambda x} dx$

$\Rightarrow$ $= -e^{-\lambda \infty} + e^{-\lambda a} = e^{-\lambda a}$

$\Rightarrow$ $\boldsymbol{E}[X] = \int_0^\infty x \lambda e^{-\lambda x} dx = \frac{1}{\lambda}$

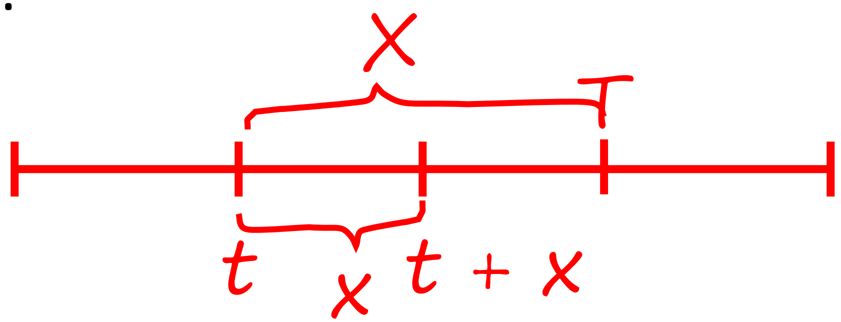
$\Rightarrow$ $\boldsymbol{E}[X^2] = \int_0^\infty x^2 \lambda e^{-\lambda x} dx = \frac{2}{\lambda^2}$

$\Rightarrow$ **var**(X) $= E[X^2] - (E[X])^2 = \frac{2}{\lambda^2} - \frac{1}{\lambda^2} = \frac{1}{\lambda^2}$

Memorylessness of the exponential PDF

- Bulb lifetime T: exponential (λ)
- Do you prefer a used or new “exponential light bulb”?

Probabilistically identical!



$\mathbf{P}$ (T > x) = $e^{-\lambda x}$, for x ≥ 0

- We are told that T > t

- r.v. X: remaining lifetime $= T - t$

Memorylessness

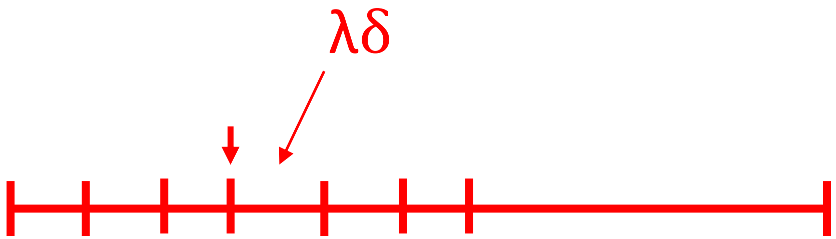
$\mathbf{P}$ (X > x | T > t) $= e^{-\lambda x}$, for x ≥ 0

$$= \frac{P(T - t > x, T > t)}{P(T > t)} = \frac{P(T > t + x, T > t)}{P(T > t)} = \frac{P(T > t + x)}{P(T > t)}$$

$$= \frac{e^{-\lambda(t+x)}}{e^{-\lambda t}} = e^{-\lambda x}$$

Memorylessness of the exponential PDF

Can human life be modelled like this?



$f_T(x) = \lambda e^{-\lambda x}$, for $x \geq 0$

$P(0 \leq T \leq \delta) \approx f_T(0) * \delta = \lambda\delta$

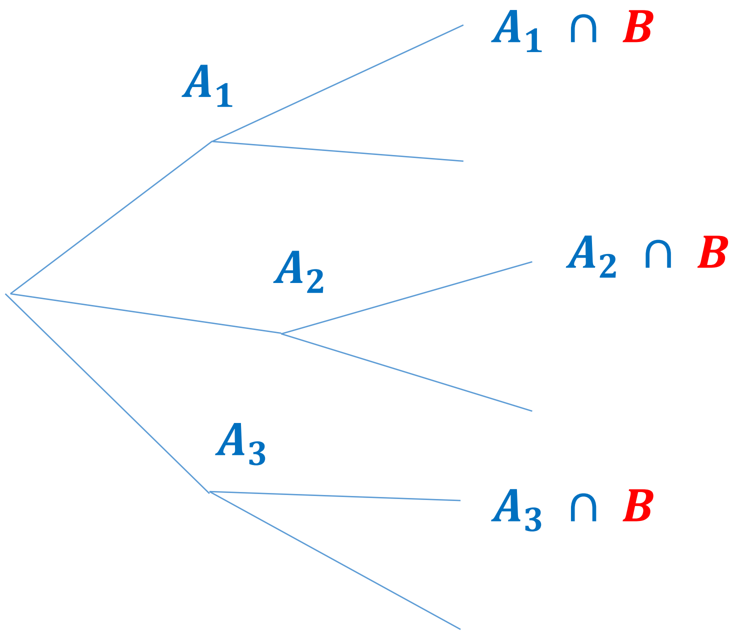
$P(t \leq T \leq t + \delta \mid T > t) = P(0 \leq T \leq \delta) \approx \lambda\delta$

similar to an independent coin flip,

Every δ time steps,

With P(success) ≈ λδ

Total probability and expectation theorems



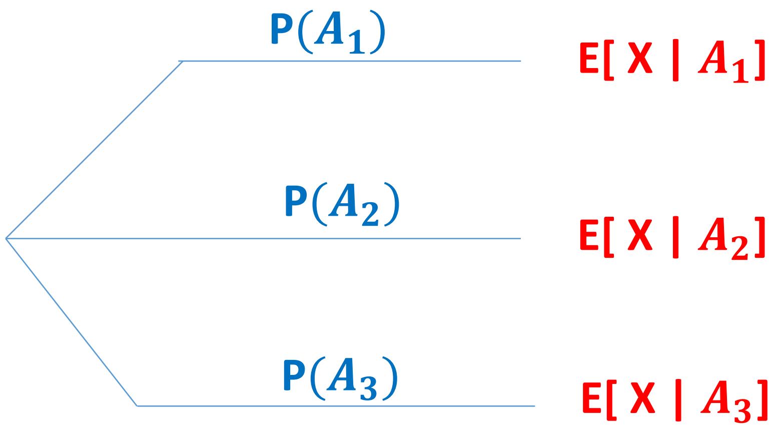
$$P(B) = P(A_1)\, P(B \mid A_1) + \ldots + P(A_n)\, P(B \mid A_n)$$

$$p_X(x) = P(A_1)\, p_{X|A_1}(x) + \ldots + P(A_n)\, p_{X|A_n}(x)$$

$$F_X(x) = P(X \le x) = P(A_1)\, P(X \le x \mid A_1) + \ldots$$

$$= P(A_1)\, F_{X|A_1}(x) + \ldots$$

$$\boxed{f_X(x) = P(A_1) f_{X|A_1}(x) + \ldots + P(A_n) f_{X|A_n}(x)}$$



$$\int x . f_X(x)\, dx = P(A_1) \int x . f_{X|A_1}(x)\, dx + \ldots$$

$$\boxed{E[X] = P(A_1) E[X|A_1] + \ldots + P(A_n) E[X|A_n]}$$

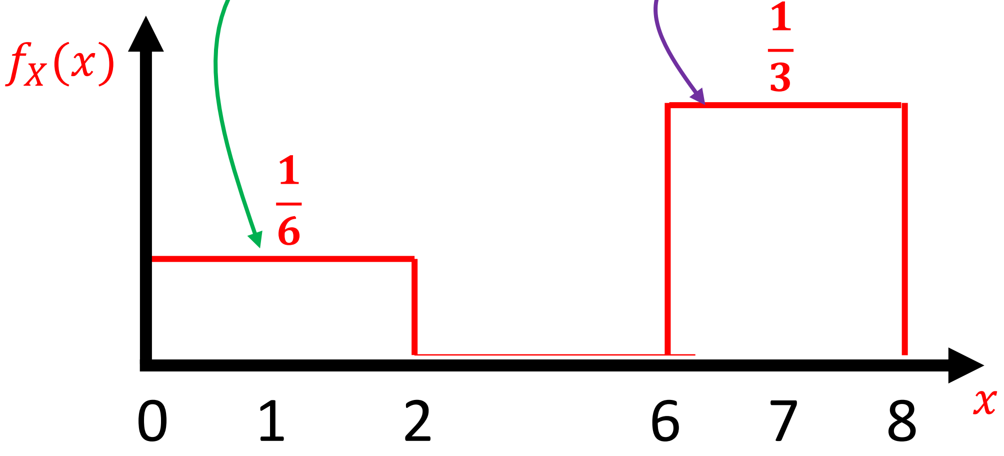
Example

Sameena goes to the supermarket shortly, with probability 1/3, at a time uniformly distributed between 0 and 2 hours from now; or with probability 2/3, later in the day, at a time uniformly distributed between 6 to 8 hours from now. Find the pdf $f_X(x)$, where RV X is the time of departure to the supermarket. Also, calculate E[X].

$f_{X|A_1} \sim U\,[0, 2]$ $\qquad$ $f_{X|A_2} \sim U\,[6, 8]$

$P(A_1) = \frac{1}{3}$ $\qquad$ $P(A_2) = \frac{2}{3}$

$$f_X(x) = P(A_1) f_{X|A_1}(x) + \ldots + P(A_n) f_{X|A_n}(x)$$



$$E[X] = P(A_1).E[X|A_1] + \ldots + P(A_n).E[X|A_1]$$

$$\frac{1}{3} * 1 + \frac{2}{3} * 7$$